PAVAN DHARMA ADAPA

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PROFESSIONAL SUMMARY

Highly motivated Computer Science graduate seeking an entry-level Data Analyst position, leveraging hands-on experience in transforming complex datasets into actionable business intelligence. Proficient in SQL, Python, R, Tableau, Microsoft Power BI, and Advanced Excel, with a track record of building end-to-end ETL pipelines, performing exploratory data analysis (EDA), and developing predictive models that improved forecast accuracy to 87% and reduced processing time by 95%. Adept at statistical analysis, KPI tracking, and data visualization, and passionate about using data storytelling to deliver clear, decision-ready insights for stakeholders in healthcare, research, and technology domains.

EXPERIENCE

Graduate Teaching Assistant
Data Science Academic Institute | Sep 2025 – May 2026

- Analyze student performance data across 6 data science courses using Advanced Excel, SQL, and Python to track KPIs such as assignment completion, grade distributions, and retention, enabling instructors to identify at-risk students early.
- Design and evaluate coding assignments that simulate real data analyst workflows, including data preprocessing, feature engineering, exploratory data analysis (EDA), and model evaluation, improving project completion speed by 25%.
- Lead lab sessions in Python (Pandas, Matplotlib) and R (ggplot2, dplyr), demonstrating how to clean, transform, and visualize datasets and communicate insights through dashboards and reports for non-technical audiences.
- Provide 1:1 mentorship on data cleansing, statistical analysis, and business-oriented analytics, helping students translate raw data into actionable insights and decision-ready summaries.

Graduate Research Analyst
Mississippi State University | Aug 2024 – Aug 2025

- Analyzed 900,000+ epidemiological records using Python, SQL, and EDA techniques to identify key environmental drivers of disease transmission, producing data-backed recommendations for research stakeholders in CSE and Veterinary Medicine.
- Built an automated ETL (Extract, Transform, Load) pipeline in Python and Selenium to collect, cleanse, and standardize multi-source datasets, reducing metadata processing time by 95% (from hours to under 10 minutes per dataset) and improving data reliability for analysis.
- Developed and evaluated ensemble predictive models (Gaussian Process Regression, Random Forest, XGBoost) to forecast disease outbreaks, achieving 87% predictive accuracy and establishing baseline KPIs for model performance tracking.
- Automated feature engineering workflows and data cleansing routines in Python, resolving 35% of dataset inconsistencies and reducing manual analysis time by 50%, which improved the quality of downstream reports and dashboards.
- Created correlation heatmaps and statistical visualizations to surface 8 key influencing factors on disease spread, packaging results into presentation-ready dashboards and reports for cross-functional faculty collaborators.
- Simulated 10,000+ outbreak scenarios using statistical analysis and hypothesis testing, translating complex modeling results into actionable insights for healthcare stakeholders and contributing to an interdisciplinary research manuscript.

Marketing Data Analyst Intern
Viral Fission | May 2023 -- Dec 2023

- Developed and optimized digital marketing analytics and reporting systems to track audience engagement, funnel performance, and campaign ROI for youth-focused brand campaigns with clients including Coca-Cola, ITC, and Mahindra.
- Built automated Power BI and GA4 dashboards that reduced manual reporting time by 50% and enabled real-time monitoring of reach, engagement rate, click-through rate, and conversion KPIs across campaigns.
- Applied SEO optimization, A/B testing, and UX insights from GA4 and social analytics to refine content and posting strategies, contributing to a 25% increase in key audience-interaction metrics over successive campaigns.
- Collaborated with brand, creative, and marketing leadership to translate event-tracking and performance data into clear, actionable recommendations on content, channel mix, and campaign timing.

Data Analytics Intern
InfraBIM Techno Solutions | Aug 2022 -- Dec 2022

- Led digital analytics and reporting-automation initiatives on engineering and operations data using Power BI, SQL, and Google Colab, improving visibility into project performance and stakeholder KPIs.
- Developed KPI dashboards, ETL pipelines, and automated reporting systems that reduced manual reporting effort by ~30% and enabled faster, data-driven decision-making for project and operations leadership.
- Conducted exploratory data analysis (EDA) to identify patterns, trends, and recurring data-quality issues, feeding findings into data-cleaning steps and feature engineering for downstream models.
- Evaluated predictive/model performance across multiple experiments, summarizing accuracy and error metrics in analysis-ready tables and visualizations to guide model selection and next-step improvements.

TECHNICAL SKILLS

- **Data Analysis & Querying:**
  SQL, Advanced Excel (Pivot Tables, VLOOKUP, lookup functions, basic automation), Python (Pandas, NumPy).

- **Business Intelligence & Data Visualization:**
  Microsoft Power BI, Tableau, Google Data Studio, Matplotlib, Seaborn; experienced in building interactive dashboards, KPI reports, and executive-ready data visualizations
- **Databases & Cloud:**
  MySQL, PostgreSQL, AWS; querying relational databases and connecting BI tools to cloud-hosted data sources
- **Statistics, Machine Learning & Experiments:**
  Statistical Analysis, Hypothesis Testing, Exploratory Data Analysis (EDA), A/B Testing, Predictive Modeling, Random Forest, XGBoost, Gaussian Process Regression, Multilayer Perceptron (MLP), Cross-Validation, Model Evaluation
- **Data Processing & Methodologies:**
  ETL (Extract, Transform, Load), Data Cleansing, Data Wrangling, Data Mining, Feature Engineering, Root Cause Analysis, KPI Tracking, Data Pipelines
- **Other Tools & Technologies:**
  Git, Linux, Selenium, Natural Language Processing (NLP), Large Language Models (LLMs), Jupyter Notebook

PROJECTS

Multimodal LLM Evaluation (LLaVA-1.5) – Analytics & Experimentation

- Designed and implemented evaluation pipelines to analyze cross-lingual performance of multimodal large language models (LLaVA-1.5), applying NLP techniques, data contamination auditing, and statistical analysis to compare model outputs across multiple benchmarks.
- Aggregated experiment metrics in Python and SQL, building clean analytical datasets and visualization-ready summaries that enabled systematic comparison of models, hyperparameters, and prompts, and supported data-driven recommendations on model selection.
- Presented findings to academic stakeholders through data storytelling, KPI-driven summaries, and visualization-rich reports, demonstrating strong stakeholder management, cross-functional collaboration, and the ability to translate complex technical results into clear, actionable insights for non-technical audiences.

REPO Sales & Marketing KPI Dashboard – Personal Data Analytics Project

- Collected and combined public sales and marketing datasets into a centralized data model using SQL and ETL (Extract, Transform, Load) pipelines, standardizing fields such as revenue, opportunities, and campaign spend to enable consistent reporting across channels.
- Built interactive sales and marketing performance dashboards in Tableau and Microsoft Power BI showcasing core business intelligence KPIs, including sales growth, sales vs. target, conversion rate by funnel stage, and product-level revenue trends, enabling at-a-glance performance monitoring for stakeholders.
- Performed exploratory data analysis (EDA), A/B-style comparisons, and root cause analysis to identify underperforming campaigns and regions, translating findings into actionable insights and data storytelling summaries for non-technical decision-makers.

EDUCATION

Master of Science in Computer Science and Engineering
Mississippi State University, Mississippi State, MS | May 2026
Relevant Coursework: Advance Machine Learning, Data Science Pedagogy, Algorithms, Artificial Intelligence, Design of Parallel Algorithms, Software Engineering

Bachelor of Technology in Computer Science and Engineering
Malla Reddy University, Hyderabad, India | May 2024
Relevant Coursework: Database Management, Statistics and Probability, Machine Learning, Data Structures, Algorithms, Data Structures

PUBLICATIONS (IN PROGRESS)

Zhang Li, Sai Harika Gade, Saikanth Ratnavale, Navicky Michael, Pavan Dharma Adapa

Predictive Modeling of Respiratory Disease Outbreaks in Poultry and Bovine Populations using Phylodynamics

- Co-authoring a healthcare analytics manuscript on predictive modeling of respiratory disease outbreaks, applying multi-host SIR frameworks to large epidemiological datasets to support data-driven public-health decisions.
- Designed and implemented the “Modeling Approaches” component, translating clinical and surveillance requirements into statistically robust multi-host SIR models and scenario analyses for healthcare stakeholders.
- Integrated pathogen and patient-level metadata to parameterize outbreak simulations, improving model realism and enabling more accurate forecasting of infection trends across animal and human populations.
- Collaborated with epidemiologists and data scientists to align simulation design, data quality checks, and reporting outputs with healthcare KPIs such as incidence rates, risk stratification, and intervention impact